Factor-combination discovery is stochastic threshold group testing: a counterfactual replay of the 2006 reprogramming screen

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Abstract

Finding the small set of transcription factors that drives a cell-fate conversion is expensive because each query is a wet-lab assay. The 2006 screen that identified the four reprogramming factors consumed a documented 61 selection experiments under a fixed leave-one-out protocol. We show that this class of screen is an instance of *stochastic threshold group testing*: a pool is negative below a lower threshold of essential factors, positive at or above an upper threshold, and stochastic in the gap between, with the gap response corresponding to the partial colony formation reported for three-of-four pools. Placing the problem in that framework, and replaying the screen on an oracle calibrated to the published assay figures, yields four results. First, information-gain design recovers the hidden recipe in a median of 10 experiments against 61, is cheaper in 300 of 300 paired replicates, and matches the known $O(g^2 d \log n)$ order for the framework; an ablation with identical inference but random pools needs 44, locating the gain in experiment selection rather than in inference. Second, the advantage is regime-specific: a classical model-free conjunctive search is twice as expensive at 24 candidates but overtakes the Bayesian designer at 48, because random pool proposals stop finding the recipe as the candidate list grows. Third, the method's only structural failure is an assumed recipe cardinality, which fails *symmetrically and silently* — zero success at 0.999 reported confidence for both smaller and larger true recipes — and is removed by treating cardinality as a latent variable at a cost of one additional experiment. Fourth, auditing the gap assumption against two measured combinatorial perturbation datasets (171 combinations, two cell systems) finds a median one-short response of 0.77 against the 0.10 the oracle assumes, placing the calibration in the error direction our own sensitivity analysis identifies as the dangerous one. We conclude that the gap response must be estimated per assay rather than assumed, and that efficient experiment selection improves cost while degrading the detectability of model error.

Author summary

When biologists search for the handful of genes that convert one cell type into another, each guess costs weeks of laboratory work. We asked a counterfactual question about a famous example: had the 2006 screen for the reprogramming factors been planned adaptively, how many experiments would it have taken? On a simulator calibrated to the original paper's numbers, about ten instead of sixty-one. More useful than that number is where the approach breaks. The problem turns out to be a known mathematical one — threshold group testing — which supplies both a theoretical yardstick and a classical algorithm as a competitor; that

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competitor wins once the candidate gene list passes about forty, so the Bayesian approach earns its complexity only in the regime the historical screen occupied. We also find a failure that matters for anyone deploying these methods: if the number of genes in the recipe is fixed in advance and that number is wrong, the search fails while reporting near-certainty, whether the true recipe is larger or smaller. Finally, we check the model's central assumption against real measurements and find it wrong in the direction that causes exactly this kind of confident failure.

1 Introduction

A cell-fate conversion driven by defined factors is, computationally, a subset-identification problem: of n candidates, an unknown subset R of size k is sufficient, and the task is to find R. What makes it hard is not the size of the space but the price of a query. Each query is a transduction experiment with a biological readout, so budgets are counted in tens.

The canonical instance is the screen of Takahashi and Yamanaka [1], extended to human cells the following year [2]. Twenty-four candidates were selected from the literature; the readout was the number of drug-resistant colonies surviving Fbx15/G418 selection; and the search followed a fixed protocol — each factor alone, the full pool, leave-one-out from the pool, then leave-one-out from the surviving subset — consuming $2n + 13 = 61$ experiments at $n = 24$ regardless of what the data said.

Adaptive experimental design [3–5] should improve on a protocol fixed in advance, and in perturbation biology several groups have shown that it does [6, 7]. What has not been established is the counterfactual for a specific documented discovery: not predictive accuracy over an adaptively collected dataset, but the number of wet-lab experiments needed to reach a known answer under the assay actually used. The 2006 screen is a good vehicle precisely because its experiment count is in the public record.

Our second contribution is a reclassification. The colony assay is not generic combinatorial search: a pool produces colonies efficiently only when it contains all of R, weakly when it contains all but one, and not at all otherwise. In group-testing language [8, 9], a test is negative at or below a lower threshold l, positive at or above an upper threshold u, and stochastic in the gap $g = u - l - 1$. This is threshold group testing [10] in its stochastic form [11], with $u = k$, $l = k - 2$ and $g = 1$. The reclassification is load-bearing rather than decorative: it supplies a theoretical yardstick for the achievable test count and classical algorithms as baselines, and both change the conclusions one would otherwise draw.

Related work. Four literatures meet here, and the closest work in each sits a measurable step away from this one.

Group testing supplies the problem class and its bounds, from the original pooled-screen argument [8] through the modern information-theoretic treatment [9]. The variant that matches this assay is threshold group testing [10] in its stochastic form [11], and designs for it continue to improve [12]. Those designs are parameterised by the threshold u: it is given, and the algorithm is built around it. Here $u = |R|$ is exactly what the experimenter does not know, which is the gap Section 2.4 addresses. Group testing is also being applied to biology right now [13], but in its disjunctive form — a pool is positive if it contains any positive item — and largely non-adaptively, whereas a reprogramming pool is positive only if it contains *all* of the recipe.

Bayesian experimental design supplies the criterion [3–5]. Applied to perturbation biology, it has been used to choose genetic interventions [6] and combination drug screens [7], and recent work learns an acquisition policy amortised over more than a thousand historical CRISPR screens [14] or

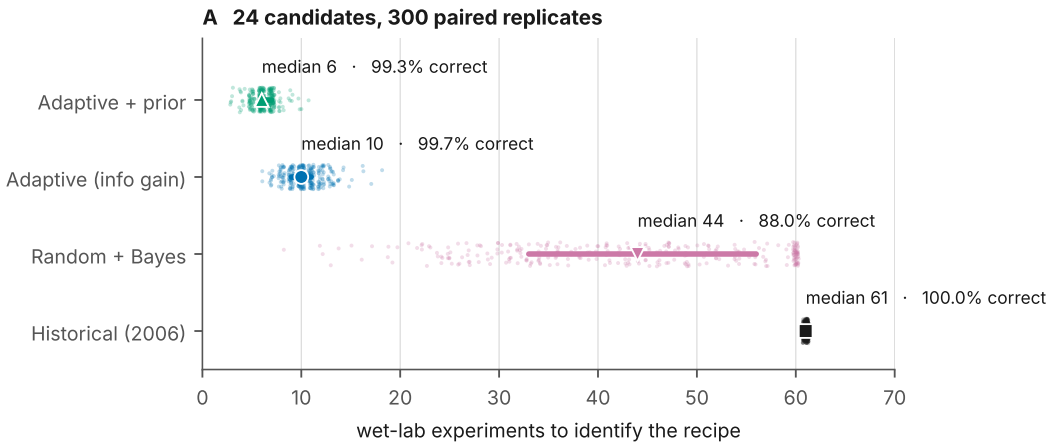


Figure 1: Wet-lab cost of four strategies at 24 candidates, 300 paired replicates. Each point is one replicate; bars span the interquartile range, markers give medians. The ablation — the same exact posterior and stopping rule with pools drawn at random — separates the contribution of experiment selection from that of inference; its pile-up at 60 is the budget ceiling.

selects gene pairs with a predictive model in the loop [15]. These are evaluated on hit-recovery or predictive accuracy at a fixed budget fraction. None asks the counterfactual question of this paper: how many experiments a documented discovery would have taken, measured against the count that discovery actually consumed.

Prediction rather than design is the third line: ranking reprogramming factors from expression and network data [16], scoring transcription-factor combinations from single-cell multiomics [17], and predicting outcomes of unseen multi-gene perturbations [18]. The closest of these even frames its accuracy in experiment-count terms — a cocktail ranked fiftieth implies fifty experiments — but the ranking is static: it does not condition on results as they arrive, and the count is a corollary of the ranking rather than a measured search cost. Such models are nonetheless the natural replacement for the oracle used here, which is the direction the virtual-cell programme points [19].

Validation of adaptive policies is the fourth. Held-out evaluations of selective measurement rules in biological screens have found that the evidence often fails to justify adopting them over a fixed plan [20], a finding that sits naturally beside Section 2.5: efficiency and error detectability pull in opposite directions.

Against that background the contributions of this paper are the counterfactual against a historical experiment count, the regime boundary against a classical algorithm, the treatment of an unknown threshold in a threshold-group-testing assay, and a check of the assay model's central assumption against measured data.

We state plainly what this study is not. The oracle is a simulator calibrated to summary figures, not an assay and not a fit to raw data; Section 2.6 is where measured data enters, and it refutes part of the calibration. No claim about cellular biology follows from any result here.

2 Results

2.1 Cost against the historical record

Over 300 replicates (Fig. 1), the historical protocol consumes its fixed 61 experiments; information-gain design needs a median of 10 (95% CI [10, 10]; mean 10.3; worst replicate 18) and recovers

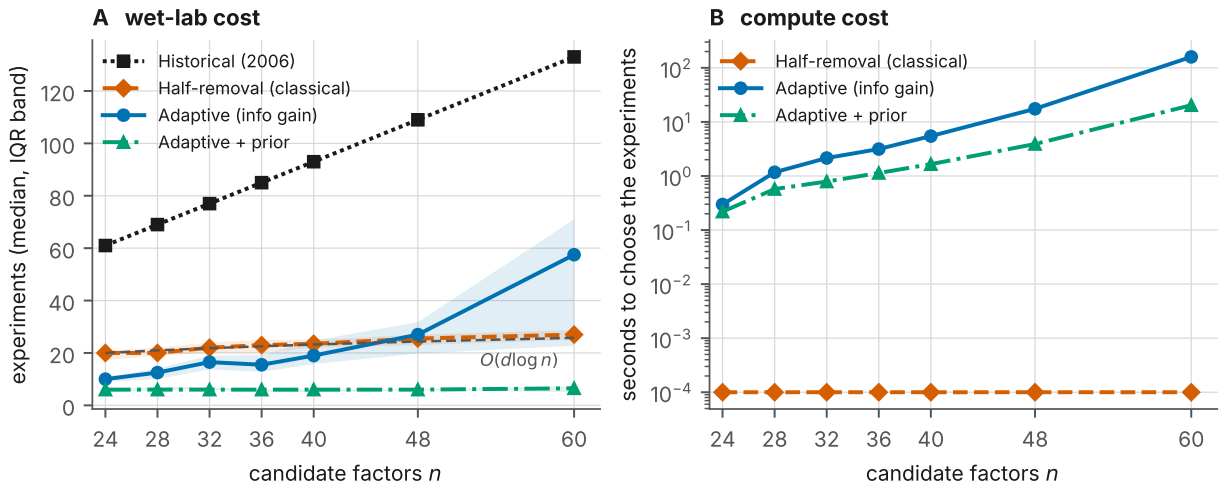


Figure 2: Growth with candidate-list size. (A) Wet-lab cost, medians with interquartile bands, against the $O(d \log n)$ reference curve scaled at the smallest list. (B) Compute spent choosing experiments, log scale. The classical half-removal search recovers logarithmic growth at negligible compute; the Bayesian designer does not, and the two cross at 48 candidates.

the exact recipe in 99.7% of runs; a literature prior reduces this to 6. Paired against the historical protocol, the adaptive designer is cheaper in 300 of 300 replicates.

Two controls matter more than the headline. The ablation needs a median of 44 experiments and recovers the recipe in 88.0% of runs; paired, the difference is 32.7 experiments (permutation $p < 10^{-4}$), with the adaptive designer cheaper in 298 of 300 replicates. The saving comes from informative experiment selection, not from Bayesian updating. The literature prior buys speed and nothing else: -4.1 experiments ($p < 10^{-4}$) with no detectable accuracy difference (McNemar $p = 1.0$).

The measured cost should be read against theory rather than against the 2006 protocol alone. For stochastic threshold group testing, test counts of order $g^2 d \log n$ are achievable [11]; at $g = 1$, $d = 4$, $n = 24$ that is ≈ 13, consistent with the observed median of 10. The 84% saving is therefore not a surprising empirical finding but the expected consequence of replacing a $\Theta(n)$ protocol with one operating at the model's known order. That is itself the point: the gap between what the 2006 screen spent and what was achievable was derivable from theory contemporaneous with it.

2.2 The advantage is regime-specific

Yamanaka's 24 candidates were themselves the product of a literature filter, so a method that works only after such filtering is of limited use. Figure 2 varies the list from 24 to 60 factors, i.e. hypothesis spaces from 10,626 to 487,635.

The Bayesian designer's wet-lab cost grows as $n^{1.8}$, not logarithmically, and its advantage over the historical protocol decays from 84% to 57%. The classical half-removal search shows the opposite pattern: twice as expensive at $n = 24$ (20 versus 10 experiments), it grows as $n^{0.4}$ and the two curves cross at $n = 48$ (26 versus 27). Because the classical search carries no model of the assay, this localises the deficiency: not in the inference and not in the problem, but in proposal generation. Random pools of size 2–12 contain all four essential factors with probability falling as n^{-4}, so as the list grows the designer evaluates information gain ever more accurately over proposals ever less likely to be informative. The remedy is a better proposal distribution, and the classical algorithm

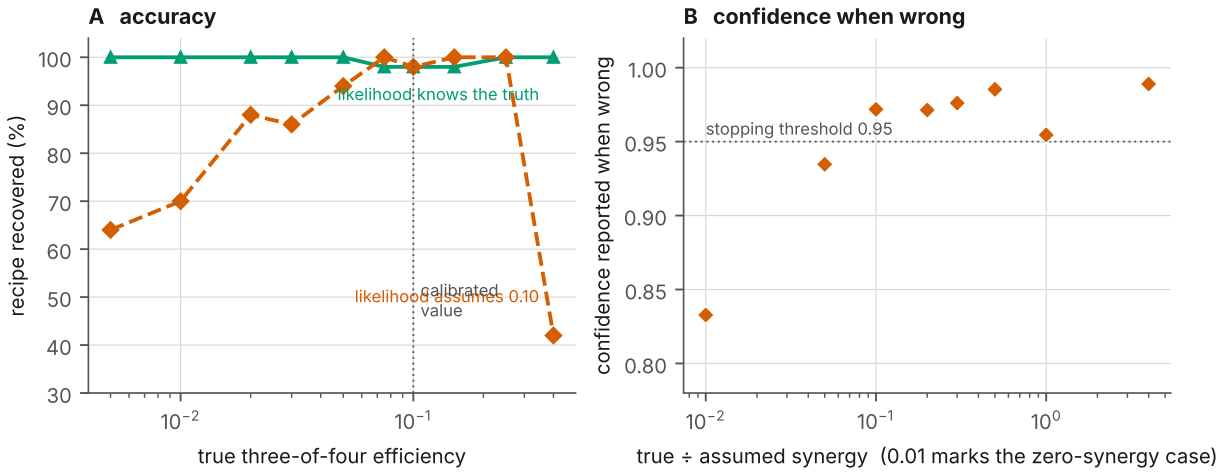


Figure 3: Gap response and calibration error. (A) Recovery rate as the true three-of-four efficiency ρ varies, with the likelihood either informed of it or holding the calibrated 0.10. (B) Confidence reported in runs that returned a wrong recipe, against the ratio of true to assumed ρ (the leftmost point is the zero-gap case).

is in effect one.

Neither method dominates. The classical search pays for its simplicity in reliability: its success rate falls from 100% to 83% as n grows to 60, for a physical reason rather than an algorithmic one. The pool-size penalty drags a full 60-factor pool down to about 6 expected colonies, so the positive control the algorithm starts from approaches the noise floor. The Bayesian designer holds 100% at every size because it reasons from the whole count distribution rather than from one threshold. Compute separates them further: 159 s per run at $n = 60$ against under a millisecond.

The $k + 1$ overlap-class collapse (Section 4.2) makes the cost of scoring a proposal independent of posterior concentration, but it remains linear in the enumerated space, and the number of rounds grows too; empirically design cost grows as $H^{1.5}$ in the hypothesis count H. Exact enumeration is the binding constraint beyond about 60 candidates. One curve in Figure 2 must be read with care: the literature-prior variant appears flat in n only because prior quality is held constant, up-weighting eight factors regardless of n. A longer candidate list implies a less informative prior, so that curve is an upper bound conditional on prior quality, not a prediction.

2.3 Model error: absent signal versus wrong signal

The graded three-of-four signal is the gap response of the threshold model, and it is natural to suppose the search depends on it. Separating the signal's presence from the searcher's belief about it shows otherwise (Fig. 3A). With the gap response removed entirely ($\rho = 0$) and the likelihood informed of this, the recipe is still recovered in 100% of runs at a median of 22 experiments: the search degenerates to pure group testing, pays a factor of 2.2 for the lost gradient, and remains 64% cheaper than the historical protocol. The same oracle with a searcher that keeps believing $\rho = 0.10$ succeeds in 52% of runs. Absence of graded synergy is a calibration error, not a structural barrier.

What binds is the magnitude of the error, asymmetrically. Under-estimating the gap response fivefold still recovers the recipe in 70% of runs, while over-estimating it fourfold drops recovery to 42% — and reports confidence 0.989 while doing so. Conservative calibration is the safer error direction, which matters because the gap response is exactly the quantity a pilot experiment estimates

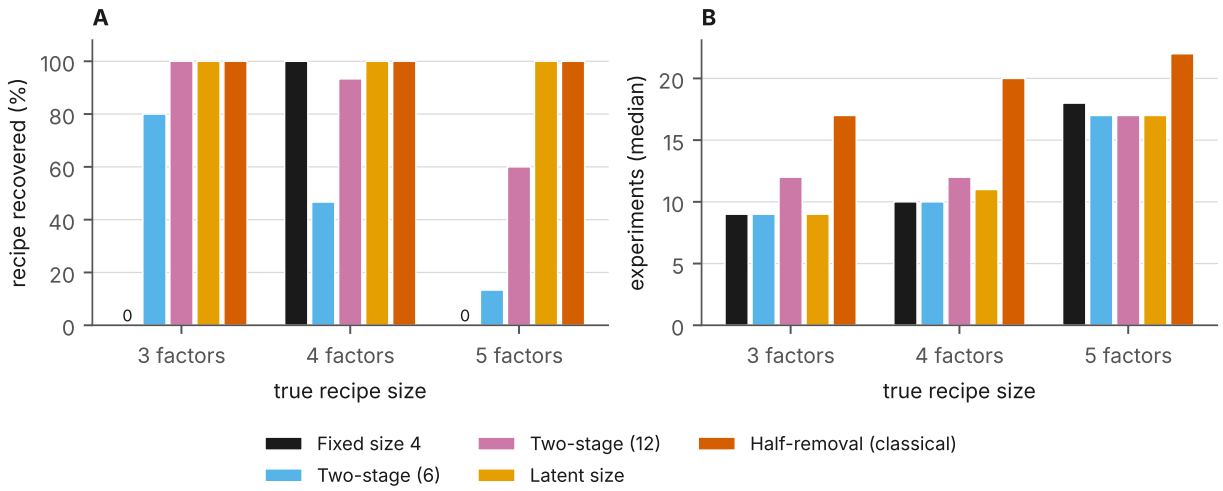


Figure 4: Unknown recipe size, several treatments, three truths. (A) Exact-recipe recovery. (B) Cost. Fixing the size at four fails completely when the truth is three or five; estimating the size in a first stage and committing to it inherits the failure whenever the estimate is wrong; carrying size as a latent variable, or dispensing with a size representation altogether, does not.

poorly.

2.4 Assumed cardinality: a symmetric, silent failure

Every result above assumes the searcher knows $k = 4$. Relaxing that assumption exposes the method's one genuinely structural failure (Fig. 4). A designer fixed at four factors recovers a three-factor recipe in 0% of runs and a five-factor recipe in 0% of runs, and in both cases reports a mean posterior confidence at or above 0.99 on the wrong answer. The failure is symmetric: searching for more factors than exist is as fatal as searching for fewer, because the fixed designer appends a spurious factor to a correct core and then concludes. This is not a failure of the data but of a hypothesis space that cannot represent the truth, and the posterior is confident precisely because every hypothesis it can represent is wrong by the same margin.

Treating cardinality as latent removes it. Recovery is 100% at true sizes three, four and five; the correct size carries posterior mass 0.995 or above in all three cases; and the premium on the four-factor case the fixed designer was built for is +0.9 experiments (paired $p = 0.0036$). The latent treatment also returns a diagnostic the fixed designer cannot produce even in principle: a posterior over recipe size, which is what would tell an experimenter that their assumed cardinality was wrong.

Two comparators place this result. The classical answer to unknown $|R|$ is to estimate the count in a first stage and then commit to it [21]. That baseline inherits the failure whenever the first stage errs, and it errs often: with a six-experiment first stage it commits to the wrong size in 53% of four-factor runs and 87% of five-factor runs, recovering the recipe in 47% and 13% of them; doubling the first stage to twelve experiments reduces the wrong commitments to 7% and 40%, for 93% and 60% recovery, at a cost of +1.7 experiments on the four-factor case ($p = 0.005$). Carrying the size posterior throughout instead of committing to it is thus not redundant machinery: it is what converts a 60% method into a 100% one at no measurable premium (+0.5 experiments, $p = 0.53$). The model-free half-removal search sidesteps the question entirely, since it never represents a size, and recovers all three sizes without modification — for +9.7 experiments ($p < 10^{-4}$), roughly double. Cardinality is therefore a liability of the parametric approach specifically, not of the problem.

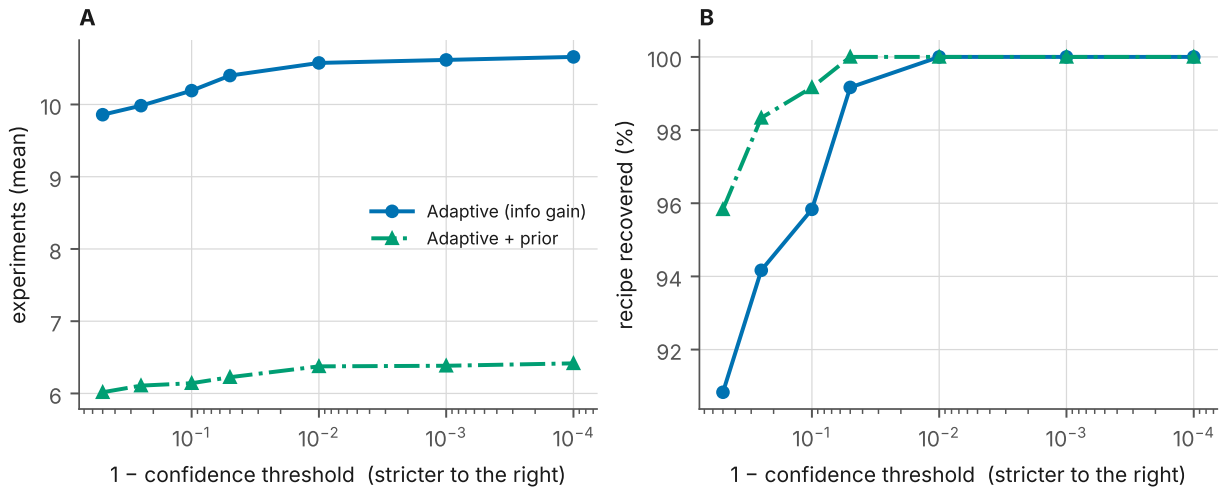


Figure 5: The price of certainty. (A) Mean experiments and (B) recovery rate as the stopping threshold tightens from 0.50 to 0.9999. Cost is nearly flat while accuracy improves, because posterior confidence jumps at the decisive experiment rather than climbing gradually.

2.5 The stopping rule, and a trade-off between efficiency and detectability

A Bayesian stopping rule is usable only if its confidence means what it says, since in a real campaign there is nothing to check it against. Sweeping the threshold from 0.50 to 0.9999 (Fig. 5), the stated confidence never exceeds the observed recovery rate by more than its Wilson interval permits (largest discrepancy +0.025), and tightening by four orders of magnitude costs 0.8 experiments on average while never moving the median. The practical recommendation is to run at 0.999 or tighter.

That flatness is the mechanism behind a pattern visible throughout. Confidence does not accumulate gradually; it jumps at the decisive experiment, so there is no intermediate regime in which the searcher can report that it is not sure yet. When the model is wrong, high confidence therefore arrives just as promptly on the wrong answer: the adaptive designer's mean confidence when wrong is 0.955 against 0.423 for the random-pool ablation; an over-estimated gap response fails at 0.989; a cardinality mismatch fails at 0.999. The designer is well calibrated *inside its model*, and that does not protect it when the model is wrong. Efficient experiment selection improves cost and degrades the detectability of model error at the same time, because a procedure that gathers consistent evidence quickly gathers it quickly for a wrong hypothesis too.

2.6 The gap assumption against measured data

Every result above is conditional on gap structure of the assumed shape existing. Single-cell perturbation screens now cover enough combinations to check it directly [24, 25], and two of them carry the pairs and singles the measurement needs. For a perturbation p with pseudobulk profile x_p and reference x_0, write $\delta_p = x_p - x_0$; for each combination whose one-short constituents were also profiled, we measure

$$\text{one-short} = \frac{\max_c \|\delta_c\|}{\|\delta_{\text{combination}}\|}, \tag{1}$$

the empirical analogue of ρ, whose calibrated value is 0.10.

In Norman et al. [22] — K562, CRISPRa, 131 pairs with both singles measured — the median is **0.77** (IQR 0.62–1.03), and **no pair of the 131** falls below 0.20. In the combinatorial subset

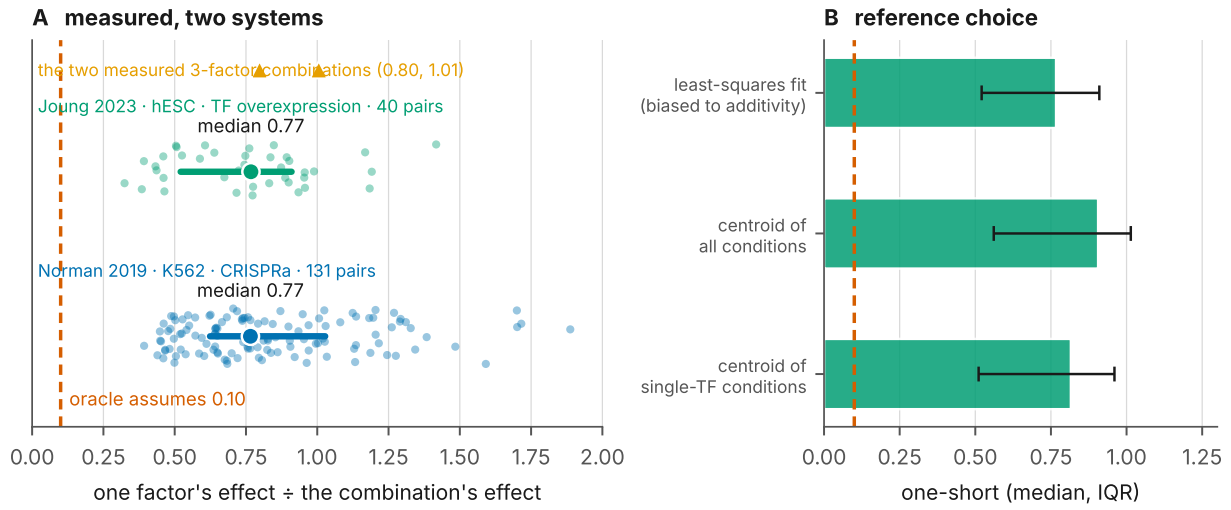


Figure 6: What one factor of a measured combination reproduces. (A) The ratio $\|\delta_{\text{one short}}\|/\|\delta_{\text{combination}}\|$ for 131 gene pairs in K562 [22] and 40 transcription-factor pairs in human embryonic stem cells [23], with the two measured three-factor combinations marked. The oracle assumes 0.10. (B) The hESC subset has no unperturbed control, so three references are reported; the conclusion does not depend on the choice.

of Joung et al. [23] — human embryonic stem cells, transcription-factor overexpression, 40 TF–TF pairs — the median is **0.77** (IQR 0.52–0.91) under a reference fitted by least squares, 0.91 and 0.82 under two alternative references, with no pair below 0.20 under any of them (Fig. 6B). The two measured three-factor combinations give 0.80 and 1.01.

The result is not an artefact of measuring effect size by a Euclidean norm. Repeating the Norman analysis with threshold-like functionals — the number of genes moving more than 0.25 or 0.5 in log space — lowers the median to 0.48 and 0.50 and raises the fraction below 0.20 to 25% and 22%, but no functional produces a median near 0.10.

Two further observations. First, departures from additivity are common but modest in magnitude: in Norman the median ratio of the pair's effect to the sum of its singles is 0.91, with 47% of pairs sub-additive and 27% super-additive, and only 28% of pairs have an additivity residual exceeding its own sampling noise, estimated by regressing squared residuals on $1/n_{AB}+1/n_A+1/n_B$. Second, the two systems differ in *where* the interaction lives: the median R^2 of a pair on its two singles is 0.86 in K562 but 0.64 in hESCs, so combinations in the stem-cell system leave the plane spanned by their constituents more, while their magnitudes remain near-additive.

The implication is specific. Our calibrated $\rho = 0.10$ is five- to eightfold sharper than measurement supports, and Section 2.3 shows that over-estimating the gap response is precisely the direction in which the designer fails silently. A designer calibrated as ours is would therefore be expected to fail confidently on these systems, while the same designer informed of the true ρ succeeds in 98–100% of runs at every value tested, including $\rho = 0$. The gap response must be estimated, not assumed.

The constructive reading is that the oracle should be replaced rather than recalibrated: a predictive model fitted to measured perturbation outcomes [18], or the broader virtual-cell programme [19], would supply a gap response estimated from data rather than assumed, and the designer would inherit whatever accuracy that model has. The experiments above then become the specification for how accurate it must be: within a factor of two on the gap response, and honest about recipe

size.

One boundary deserves emphasis. Both datasets read out the transcriptome, while the 2006 assay reads out a rare discrete phenotype — a colony forms or does not — which is a thresholded functional of cell state and may well be far sharper than transcriptome displacement. The audit therefore refutes the assumption for expression-level objectives and constrains, rather than settles, the phenotype case. Settling it requires a combinatorial screen with a phenotypic readout, which to our knowledge does not exist at the scale needed.

3 Discussion

Four conclusions survive the controls, and they are narrower than the headline saving.

The saving is real but unsurprising once the problem is named. Recognising the colony assay as stochastic threshold group testing with $g = 1$ supplies the achievable order, $O(g^2 d \log n)$, and the observation that the 2006 protocol operated at $\Theta(n)$. The contribution is not the adaptive designer but the quantification: how much of that theoretical gap is realisable under the assay's noise, with the historical count as the reference.

The Bayesian apparatus earns its place only in a regime. At the scale the historical screen occupied it is twice as efficient as a classical conjunctive search; beyond 48 candidates it is worse in experiments and five orders of magnitude worse in compute, while remaining more reliable. Any application to larger candidate lists should either repair the proposal distribution, plausibly by borrowing the classical algorithm's structure, or use the classical algorithm with the Bayesian machinery reserved for the final discrimination.

For deployment, the cardinality result is the most actionable: a parametric designer that fixes recipe size fails silently and symmetrically, the standard two-stage remedy inherits the failure, and latent-cardinality inference removes it for about one extra experiment while supplying a size posterior that makes the error visible. Given the general pattern that efficient design suppresses the signature of model error, cheap diagnostics of this kind are not optional extras.

Finally, the real-data audit turns the study's main caveat into a result. The assumption that a one-short pool yields a tenth of the full response is not supported in either measured system; the measured value is close to 0.8. Since over-estimating that response is the error direction that produces confident failure, the operational rule follows directly: measure the gap response in a pilot, calibrate conservatively, and treat a designer that has not done so as untrustworthy regardless of what its posterior reports.

Limitations. The oracle is a simulator calibrated to published summary figures, so its numbers are properties of a stated model; the design of the comparisons, not the realism of the oracle, carries the conclusions. The audit datasets read out expression rather than a rare phenotype, which is the readout the historical assay used; the evaluator-quality lesson of Okita et al. [26], where a better selection marker changed which colonies counted as success, is the biological form of the same concern. The historical protocol is reconstructed from the paper's description rather than laboratory records. The latent-cardinality designer widens its assumption to sizes 3–6 rather than eliminating it, so a seven-factor truth would reproduce the original failure. Exact enumeration limits the approach to candidate lists of order 10^2, which excludes the genome-scale setting where a reprogramming screen would benefit most.

4 Methods

4.1 Oracle

A hidden essential set $R \subseteq \{1, \ldots, n\}$, $|R| = k$, is drawn uniformly. For a transduced pool S, the colony count is a Poisson draw,

$$y(S) \sim \text{Poisson}(\lambda(S)), \qquad \lambda(S) = B\,\phi(|S \cap R|)\exp(-\delta \max(0, |S| - k)), \tag{2}$$

with $\phi(k) = 1$, $\phi(k-1) = \rho$ and $\phi(j) = 0$ for $j < k-1$. Calibration targets from Takahashi and Yamanaka [1]: a clean four-factor pool yields about 45 colonies ($B = 45$); a three-of-four pool yields a small number lacking full ES-like morphology ($\rho = 0.10$, about 4 colonies); per-cell efficiency falls as more factors are transduced, which is why the reported 10-factor subset outperformed the full 24 ($\delta = 0.0357$, giving 36 and 22 colonies). In threshold-group-testing terms, $u = k$, $l = k-2$, $g = 1$, and ρ is the gap response.

4.2 Inference

The hypothesis space is every k-subset of the candidates ($\binom{24}{4} = 10{,}626$ in the calibrated case), small enough for an exact posterior with no sampling. The next pool maximises expected information gain [3], $\text{EIG}(S) = H[p(y \mid S)] - \mathbb{E}_h\, H[p(y \mid S, h)]$, over 120–260 random pools of size 2–12 plus pools assembled from the most probable factors. The count distribution depends on a hypothesis only through $|S \cap h|$, which takes $k+1$ values, so likelihood and information gain collapse onto $k+1$ overlap classes and the cost of scoring a proposal is independent of posterior concentration. The search stops when the leading hypothesis exceeds τ (0.95 unless stated) or the budget is exhausted.

For unknown cardinality the space becomes every subset of size 3–6 (189,750 at $n = 24$) and the rate depends on a hypothesis through the pair $(|h|, |S \cap h|)$, giving 22 reachable classes, so the same collapse applies one dimension up.

4.3 Strategies and baselines

Historical protocol. n single-factor experiments, the full pool, n leave-one-out experiments, the surviving 10-factor pool, 10 leave-one-out experiments, and a confirmation: $2n + 13$, non-adaptive. **Adaptive design**, with a uniform prior, and **with a literature prior** up-weighting eight factors (the true ones plus decoys), whose quality is varied over coverage, breadth and strength in a separate experiment. **Random pools with Bayesian updating** is the ablation: identical inference, pools drawn at random. **Half-removal search** is the conjunctive analogue of binary splitting, carrying no model beyond a positivity threshold and no representation of recipe size: half the undecided factors are removed and the pool retested; if it remains positive every removed factor was inessential and the candidate set halves, and if it turns negative the removed half contains an essential factor, found by binary search. Its one hyperparameter, the positivity threshold, is the geometric mean of the assumed full and gap rates at the pool size tested, so it carries the same calibration error as the Bayesian designer. **Two-stage cardinality estimation** spends a fixed budget (6 or 12 experiments) estimating the count, then commits to it [21]. **Latent-cardinality design** carries size and identity in one posterior.

4.4 Campaign and statistics

Results come from independent replicates recorded individually, so that all aggregation is reproducible without re-simulation; strategies within an experiment share hidden recipes and noise seeds,

making every comparison paired. Cost is summarised by medians with interquartile ranges, because experiment counts are integer, bounded below and right-skewed. Median intervals are percentile bootstrap (20,000 resamples); proportions use Wilson score intervals, which retain width at 0% and 100%. Paired cost comparisons use a two-sided paired permutation test on the mean difference (20,000 sign flips); paired accuracy comparisons use an exact McNemar test. Success thresholds are estimated by logistic regression in $\log_{10}$ of the varied parameter and reported only when the crossing lies inside the tested range. The statistics layer is implemented in NumPy and checked against closed-form or planted-truth answers, including the permutation test's false-positive rate under the null.

4.5 Measured datasets

Norman et al. [22] was obtained from the scPerturb harmonisation [25] (`NormanWeissman2019_filtered.h5ad`) and Joung et al. [23] from its combinatorial subset. Counts were normalised to 10^4 per cell, log1p transformed, restricted to the 2000 most variable genes, and averaged per perturbation over conditions with at least 20 cells, giving 237 and 52 conditions respectively. In the hESC subset GFP is an inert partner, so `X+GFP` is the single for X; no GFP-only condition was profiled, so three references are reported, including one fitted by least squares that is deliberately biased toward additivity.

Data and code availability

All code, the per-run records, the analysis layer and its tests, and scripts regenerating every figure are available at `https://github.com/OWNER/yamanaka-replay` under the MIT licence. The headline comparison reproduces in under a minute on two CPU cores with NumPy as the only dependency; the full campaign takes approximately two hours. The two measured datasets are public and the preparation script is included.

Acknowledgements

The code and analyses were implemented with the assistance of Claude (Anthropic) under the direction of the author, who designed the study, specified the experiments and is responsible for all claims and interpretations.

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